

HisabDo AI Assistant

Integration Planning Documentation

Purpose: This document explains, in simple words, how the HisabDo AI Assistant will connect with the HisabDo website, backend, AI service, and future mobile app.

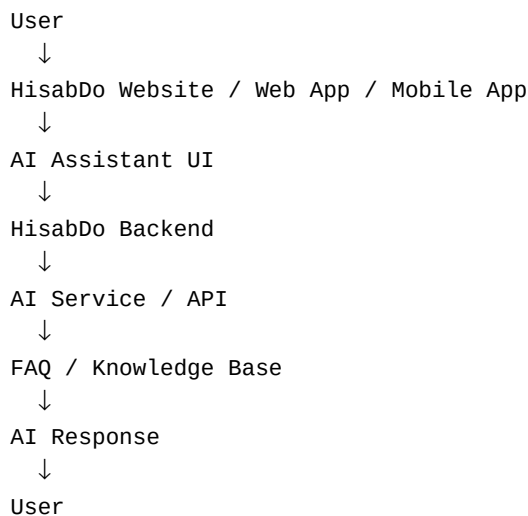
1. What is Integration Planning?

Integration planning means deciding how the AI Assistant will connect with the existing HisabDo system. The goal is to make sure that a user can ask a question in the HisabDo application, the question reaches the AI service, the AI finds the correct information from the FAQ or knowledge base, and the answer comes back to the user.

Simple example: A user asks, "How can I create an invoice?" The HisabDo application sends the question to the backend. The backend sends it to the AI service. The AI service searches the HisabDo knowledge base and returns a helpful answer.

2. Integration Architecture

The following design shows the complete path of a user question:



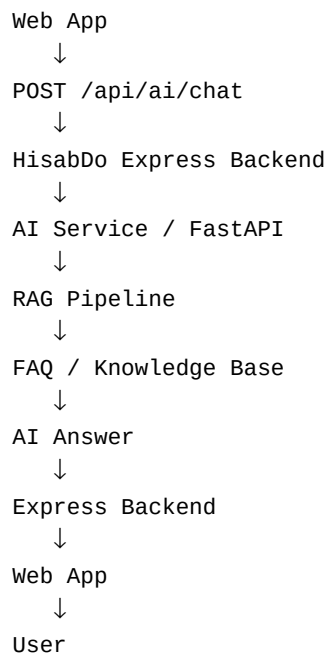
3. How the Architecture Works

Part	Simple Explanation
1. User	The user asks a question using the AI Assistant.
2. Website / Web App / Mobile App	The application provides the screen where the user can open and use the chatbot.
3. AI Assistant UI	This is the chat window, input box, send button, messages, loading indicator, and error messages.
4. HisabDo Backend	The backend receives the user's request, checks authentication, validates the request, and sends it to the AI service.

5. AI Service / API	The AI service processes the question and uses the RAG system to find useful information.
6. FAQ / Knowledge Base	This contains HisabDo information such as features, FAQs, and user guidance.
7. AI Response	The AI creates an answer and sends it back through the backend to the application.
8. User	The user sees the answer inside the AI Assistant.

4. API Communication Flow

The frontend should communicate with the HisabDo backend. The backend then communicates with the AI service. This keeps the AI service and secret keys protected.



5. Example API Request and Response

User Question: How can I create an invoice?

Request:

```
{
  "message": "How can I create an invoice?"
}
```

Response:

```
{
  "answer": "You can create an invoice from the Invoices section."
}
```

6. Website Integration Plan

The AI Assistant can be added to the HisabDo website or web application as a chatbot button or chat panel. The frontend will call the HisabDo backend when the user sends a question.

- Add an AI Assistant button or chat icon.
- Create the chat window.

- Add a text box and Send button.
- Show user and AI messages.
- Show a loading message while the AI is working.
- Show a simple error message if the request fails.
- Connect the frontend to the AI API.

7. Frontend Integration Points

Frontend integration points are the places in the web application that need changes for the chatbot.

- AI Assistant component
- Chat window
- Message list
- User input box
- Send button
- API service
- Loading state
- Error handling
- Authentication token handling

8. Backend Integration Points

The HisabDo backend acts as the middle layer between the application and the AI service.

- Create an AI chat API endpoint such as POST /api/ai/chat.
- Receive the user's question.
- Check that the user is authenticated.
- Validate the request.
- Send the question to the AI service.
- Receive the AI answer.
- Return the answer to the frontend.

9. Web App Integration Plan

The current web application can use the same HisabDo backend and AI service. The web application does not need to contain the AI model itself. It only needs to communicate with the backend API.

Web App → HisabDo Backend → AI Service → Knowledge Base → Answer

10. Mobile App Integration Plan

If HisabDo gets a mobile application in the future, the mobile app can use the same backend and AI service. There is no need to create a separate AI system for mobile.

Mobile App → HisabDo Backend → AI Service → Knowledge Base → Answer

This makes the system easier to maintain because the web app and mobile app can use the same AI logic.

11. Authentication and Security

Security means protecting the AI API, user accounts, and private information.

- Require users to be authenticated before using protected AI features.
- Verify the user's authentication token on the backend.
- Keep AI and other secret API keys on the server, never in frontend code.
- Use HTTPS in production.
- Validate user input before processing it.
- Add rate limiting to reduce abuse and excessive requests.
- Do not expose internal server details or secret information in error messages.
- Keep logs safe and avoid storing sensitive information unnecessarily.

12. Data Privacy

HisabDo may contain business and financial information. The AI Assistant should receive only the information it needs.

- Do not send unnecessary private or financial data to the AI service.
- Never send passwords or secret credentials to the AI model.
- Use minimum required data for each request.
- Protect stored conversations if conversation history is added.
- Control who can access AI-related data.

13. Future Scalability

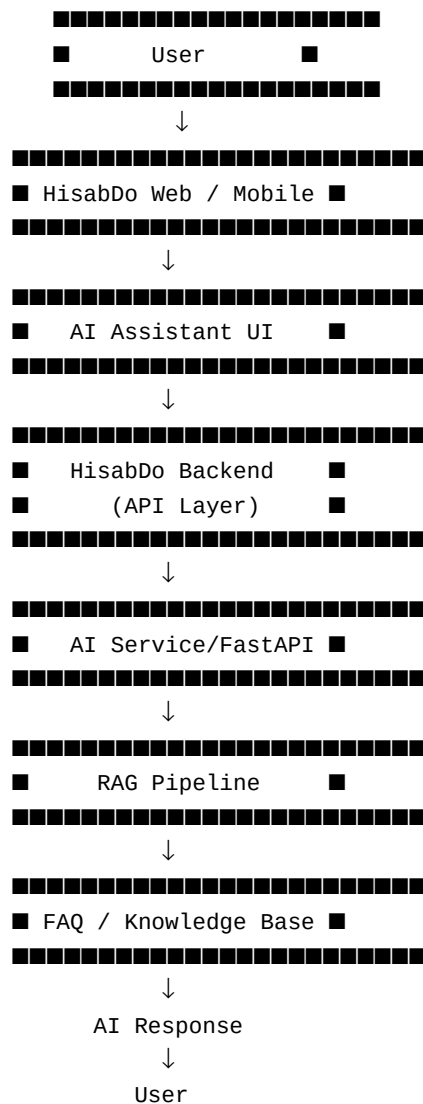
Scalability means making sure the system can handle more users and more questions in the future.

- Use a separate AI service so it can be scaled independently.
- Add caching where appropriate.
- Use rate limiting.
- Add load balancing when traffic becomes high.
- Run multiple AI service instances when needed.
- Store conversation history in a suitable database if required.
- Add monitoring and logging to find problems quickly.

14. Complete Request and Response Example

1. User opens the HisabDo AI Assistant.
2. User asks: "How do I add a customer?"
3. Frontend sends the question to the HisabDo backend.
4. Backend checks the user and validates the request.
5. Backend sends the question to the AI service.
6. AI service searches the HisabDo FAQ / knowledge base.
7. RAG finds the most relevant information.
8. AI creates the answer.
9. AI service sends the answer to the backend.
10. Backend sends the answer to the frontend.
11. Frontend displays the answer to the user.

15. Final Integration Design



16. Expected Output

- Integration Architecture
- API Communication Flow
- Website Integration Plan

- Web App Integration Plan
- Mobile App Integration Plan
- Frontend Integration Points
- Backend Integration Points
- Security Considerations
- Data Privacy Plan
- Future Scalability Plan

17. Simple Summary

The main idea: The HisabDo application shows the AI Assistant to the user. The HisabDo backend receives the question and safely sends it to the AI service. The AI service uses the RAG system and HisabDo FAQ/knowledge base to find the correct information. The answer then returns through the backend and is displayed to the user. The same backend and AI service can later be used by a HisabDo mobile app.